```
yuxuan@yuxuan-Alienware-x15-R1:~/Desktop/cs454-midterm-starter$ multipass info cs554-midterm
Name:           cs554-midterm
State:          Running
Snapshots:      0
IPv4:           10.213.186.117
Release:        Ubuntu 24.04.5 LTS
Image hash:     04e8f2461a16 (Ubuntu 24.04 LTS)
CPU(s):         4
Load:           0.00 0.00 0.00
Disk usage:     1.9GiB out of 29.0GiB
Memory usage:   407.6MiB out of 3.8GiB
Mounts:         --
yuxuan@yuxuan-Alienware-x15-R1:~/Desktop/cs454-midterm-starter$ multipass shell cs554-midterm
Welcome to Ubuntu 24.04.5 LTS (GNU/Linux 6.8.0-147-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Sun Oct 18 20:35:42 CDT 2026

  System load:  0.0               Processes:             124
  Usage of /:   6.7% of 28.02GB   Users logged in:       0
  Memory usage: 6%                IPv4 address for ens3: 10.213.186.117
  Swap usage:   0%


Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

New release '26.04.1 LTS' available.
Run 'do-release-upgrade' to upgrade to it.


To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.
```

```
ubuntu@cs554-midterm:~/cs554-midterm$ sudo docker compose up -d
[+] up 2/2
 ✔ Network cs554-midterm_default Created
 ✔ Container cs554-midterm-web-1 Started
ubuntu@cs554-midterm:~/cs554-midterm$ sudo docker compose ps
NAME                IMAGE             COMMAND                  SERVICE   CREATED         STATUS         PORTS
cs554-midterm-web-1   cs554-midterm:v1   "docker-entrypoint.s…"   web       7 seconds ago   Up 7 seconds   0.0.0.0:8080->8080/tcp, [::]:8080->8080/tcp
ubuntu@cs554-midterm:~/cs554-midterm$ curl -i -w '\n' http://localhost:8080/
HTTP/1.1 200 OK
Content-Type: application/json
Date: Mon, 19 Oct 2026 02:52:56 GMT
Connection: keep-alive
Keep-Alive: timeout=5
Transfer-Encoding: chunked

{"service":"cs554-midterm-compose","message":"Hello from Docker Compose inside the VM!","hostname":"95b02f2d2fe6","requestCount":1}
ubuntu@cs554-midterm:~/cs554-midterm$ curl -i -w '\n' http://localhost:8080/health
HTTP/1.1 200 OK
Content-Type: application/json
Date: Mon, 19 Oct 2026 02:53:17 GMT
Connection: keep-alive
Keep-Alive: timeout=5
Transfer-Encoding: chunked

{"status":"ok"}
ubuntu@cs554-midterm:~/cs554-midterm$ sudo docker compose logs --no-color web
web-1  | Listening on port 8080
web-1  | {"method":"GET","path":"/","status":200}
web-1  | {"method":"GET","path":"/health","status":200}
ubuntu@cs554-midterm:~/cs554-midterm$ exit
logout
yuxuan@yuxuan-Alienware-x15-R1:~/Desktop/cs454-midterm-starter$
```

```
yuxuan@yuxuan-Alienware-x15-R1:~/Desktop/cs454-midterm-starter$ multipass info cs554-midterm
Name:           cs554-midterm
State:          Running
Snapshots:      0
IPv4:           10.213.186.117
                172.17.0.1
                172.18.0.1
Release:        Ubuntu 24.04.5 LTS
Image hash:     04e8f2461a16 (Ubuntu 24.04 LTS)
CPU(s):         4
Load:           0.00 0.00 0.00
Disk usage:     3.0GiB out of 29.0GiB
Memory usage:   509.9MiB out of 3.8GiB
Mounts:         --
yuxuan@yuxuan-Alienware-x15-R1:~/Desktop/cs454-midterm-starter$ curl -i -w '\n' http://10.213.186.117:8080/
HTTP/1.1 200 OK
Content-Type: application/json
Date: Mon, 19 Oct 2026 03:00:05 GMT
Connection: keep-alive
Keep-Alive: timeout=5
Transfer-Encoding: chunked

{"service":"cs554-midterm-compose","message":"Hello from Docker Compose inside the VM!","hostname":"95b02f2d2fe6","requestCount":2}
```

```
ubuntu@cs554-midterm:~$ cd ~/cs554-midterm
ubuntu@cs554-midterm:~/cs554-midterm$ curl -i -w '\n' http://localhost:8080/
HTTP/1.1 200 OK
Content-Type: application/json
Date: Mon, 19 Oct 2026 03:08:18 GMT
Connection: keep-alive
Keep-Alive: timeout=5
Transfer-Encoding: chunked

{"service":"cs554-midterm-compose","message":"Hello from Docker Compose inside the VM!","hostname":"95b02f2d2fe6","requestCount":3}
ubuntu@cs554-midterm:~/cs554-midterm$ sudo docker compose restart web
[+] restart 0/1
 ⠸ Container cs554-midterm-web-1 Restarting
ubuntu@cs554-midterm:~/cs554-midterm$ curl -i -w '\n' http://localhost:8080/
HTTP/1.1 200 OK
Content-Type: application/json
Date: Mon, 19 Oct 2026 03:09:22 GMT
Connection: keep-alive
Keep-Alive: timeout=5
Transfer-Encoding: chunked

{"service":"cs554-midterm-compose","message":"Hello from Docker Compose inside the VM!","hostname":"95b02f2d2fe6","requestCount":1}
```